

Program overview

19-May-2025 14:40

Year 2024/2025
 Organization Industrial Design Engineering
 Education Minors Industrial Design Engineering

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Minors Industrieel Ontwerpen	Minors Industrial Design Engineering		
IO-MI-075	IO-MI-075-24 Study Abroad		
IO3891	Report Minor Study Abroad (1 EC)	1	
IO3892	Report Minor Study Abroad (2 EC)	2	
IO3893	Report Minor Study Abroad (3 EC)	3	
In het buitenland te behalen vakken, minimaal 21 EC			
In binnen- of buitenland te behalen vakken, minimaal 6-9 EC			
IO-MI-076	IO-MI-076-24 Workplacement		
IO3820-15	Internship	30	
IO-MI-124	IO-MI-124-24 Interactive Environment		
IO-MI-221	IO-MI-221-24 Advanced Prototyping		
IO-MI-222	IO-MI-222-24 Designing Sustainability Transitions		
IO3834-23	Value Sensitive Design	4	
IO-MI-223	IO-MI-223-24 Connected Creativity		
IO-MI-240	IO-MI-240-24 Future Mobility Design		

Year	2024/2025
Organization	Industrial Design Engineering
Education	Minors Industrial Design Engineering

Minors Industrieel Ontwerpen

Year	2024/2025
Organization	Industrial Design Engineering
Education	Minors Industrial Design Engineering

IO-MI-075

IO3891	Report Minor Study Abroad (1 EC)	1
Course Coordinator	Ir. E.J.J. van Breemen	
Course Contents	Additional report type 1 for extra credits	
Study Goals	See Minor Study Abroad	
	See website for further information: https://www.tudelft.nl/studenten/io/onderwijs/study-abroadexchange	
Education Method	Generating report based on reflection and contemplation	
Assessment	Report	

IO3892	Report Minor Study Abroad (2 EC)	2
Course Coordinator	Ir. E.J.J. van Breemen	
Course Contents	Additional report type 2 for extra credits	
Study Goals	See Minor Study Abroad	
	See website for further information: https://www.tudelft.nl/studenten/io/onderwijs/study-abroadexchange	
Education Method	Generating report based on reflection and contemplation	
Assessment	Report	

IO3893	Report Minor Study Abroad (3 EC)	3
Course Coordinator	Ir. E.J.J. van Breemen	
Course Contents	Additional report type 3 for extra credits	
Study Goals	See Minor Study Abroad	
	See website for further information: https://www.tudelft.nl/studenten/io/onderwijs/study-abroadexchange	
Education Method	Generating report based on reflection and contemplation	
Assessment	Report	

Year	2024/2025
Organization	Industrial Design Engineering
Education	Minors Industrial Design Engineering

In het buitenland te behalen vakken, minimaal 21 EC

Year	2024/2025
Organization	Industrial Design Engineering
Education	Minors Industrial Design Engineering

In binnen- of buitenland te behalen vakken, minimaal 6-9 EC

Year	2024/2025
Organization	Industrial Design Engineering
Education	Minors Industrial Design Engineering

IO-MI-076

IO3820-15	Internship	30
Course Coordinator	J.J.A.P.M. van Domburg	
Education Period	None (Self Study)	
Exam Period	none	

Year	2024/2025
Organization	Industrial Design Engineering
Education	Minors Industrial Design Engineering

IO-MI-124	
Program Coordinator	Ing. A.J.C. van der Helm
Program Title	Interactive Environments Minor
Contact for Students	Aadjan van der Helm - a.j.c.vanderhelm@tudelft.nl
Minor Remarks / Schedule	For more information about this minor see the following link: https://interactive-environments.nl/ This is a collaboration course. External partners bring in real life cases and obtain IP rights to the results. At the start of the course the course staff will ask you to sign the IP(R) transfer with TU Delft to make collaboration possible. More detailed information can be found in the student portal -> practical affairs -> Intellectual Property in IDE education -> collaboration courses.

Year	2024/2025
Organization	Industrial Design Engineering
Education	Minors Industrial Design Engineering

IO-MI-221

Responsible Program Employee	W.S. Elkhuisen
Contact for students	prototyping@tudelft.nl
EC Program	30
Program Start (Study Year)	2019
Program Goals	The Advanced Prototyping Minor brings together the most recent digital manufacturing methods and virtual/mixed reality technology, in creating functional high-fidelity prototypes. Prototypes are used in various design and engineering disciplines as an early version of a product or a system. They are built for a variety of purposes including to test an idea or concept (e.g., proof-of-concept prototypes), to represent certain aspects of an intended design (e.g., visual/ functional prototypes), and to trigger discussion in a multi-disciplinary design team (e.g., boundary objects). The course aims at equipping you with a wide range of advanced prototyping skills, techniques and materials to create prototypes, which capture both function and appearance of the intended design. To that aim, the course covers a broad interdisciplinary foundation, in-depth knowledge and practical skills, through (guest) lectures, hands-on workshops and multidisciplinary research/design projects. In addition to the knowledge and skills necessary for building prototypes, the course injects necessary critical and creative thinking to determine which of these techniques and when in the design process could be applied, by considering desirability, feasibility, viability, as well as sustainability.
Administration by the Faculty of	IO
Intended for	Advanced Prototyping Minor suits both design-oriented students with science/technology background and science/technology oriented students with art/design background. The aim is to build project groups by combining students from these different backgrounds to create an environment in which students can reflect on the contribution of their own discipline for advanced prototyping. Selection takes place by means of a draw, based on the following formula: 1/2 from Industrial Design (Engineering) 1/2 from other (engineering) disciplines This allocation is done to ensure the interdisciplinary nature of the minor. We welcome (international) exchange students. Minors of the faculty of Industrial Design Engineering are open only for students from academic programmes.
Minor Content 1	The Advanced Prototyping Minor consists of two courses: - Prototyping with/for Digital Fabrication (15EC, Q1) - Advanced Prototyping Project (15EC, Q2) Q1 In the course 'Prototyping with/for Digital Fabrication' (PDF), you will learn about advanced prototyping, in relation to 3D digitisation, generative design, digital fabrication and extended reality (VR/AR). The course includes the following 5 modules: - Low-fidelity prototyping - Digitisation - Design automation - Digital fabrication - Extended reality prototyping In the 'low-fidelity prototyping' module, you will learn different techniques for rapid prototyping (e.g. with cardboard, clay, paper, etc.), which you can apply in early design phases, and for later rapid iterations, to evaluate design ideas quickly and efficiently. In the 'digitisation' module, you will learn about and apply various 3D scanning technologies and other data capture technologies (e.g., pressure measurement) to create a digital twin of people/objects. In the 'design automation' module, you will learn parametric design (e.g., with Rhino Grasshopper), and learn about topics such as topology optimization and generative design principles. In the 'digital manufacturing' module, you will learn about and use additive and subtractive digital manufacturing technologies (e.g., 3D printing, laser cutting, CNC milling, CNC turning, waterjet cutting, etc.). Finally, in the 'extended reality prototyping' module, you will learn to use VR/AR technology at different stages of the design process: for design creation (e.g. VR sketches), for design evaluation (e.g. user testing using VR/AR), and for creating virtual and/or mixed reality applications. Q2 In the Advanced Prototyping Project course, you will apply the knowledge and skills gained from Q1 in a group project. An iterative design process will be followed, in which you will design and build (multiple) prototype(s). Besides building prototypes, you will test and evaluate the prototypes and eventually exhibit both the process and the prototypes in a public location. In Q2 you will also attend several workshops on transferable skills, such as (prototype) photography, video making and pitching. In addition, there are several workshops on material testing and user testing of prototypes. What's in it for you? - Lectures and guided workshops on various 3D scanning, parametric modeling, digital fabrication technologies, and virtual/mixed reality prototyping - Collaboration with other disciplines, including engineering, science, art and design. - Access to and use of facilities/labs, including Makerspace in the TU Delft Science Center, prototyping and modeling workshop (PMB@IDE), the Body Lab (digitization @IDE), and TU Delft VR Zone.
Minor Assessment	Both courses in the minor are assessed through either individual or group project assignment. These will be graded using a rubric. The grading of the assignments will be conducted by lecturers/tutors of the course. Benchmarking will take place in the case of multiple, parallel coaches to ensure consistency. Every individual course will have to be passed with at least 6/10.
Maximum number of participants	45
Minor Remarks / Schedule	This is a collaboration course. External partners bring in real life cases and obtain IP rights to the results. At the start of the course the course staff will ask you to sign the IP(R) transfer with TU Delft to make collaboration possible. More detailed information can be found in the student portal -> practical affairs -> Intellectual Property in IDE education -> collaboration courses.

Year	2024/2025
Organization	Industrial Design Engineering
Education	Minors Industrial Design Engineering

IO-MI-222	
Minor Coordinator	Ir. C.P.J.M. Kroon
Contact for students	Ir. Caroline Kroon - c.p.j.m.kroon@tudelft.nl
Program Title	Designing Sustainability Transitions
In association with the Faculty of	IO
EC Program	30
Program Start (Study Year)	2019
Program Goals	The minor is characterised by a dual approach to sustainability transitions. The courses stimulate high-level critical thinking through lectures, talks by external experts and group discussions. On the other hand stimulating the development of practical design skills for sustainability through design courses, design masterclasses, and working in multi-disciplinary student teams on the integral design challenge. The learning goals are: - Analyse the different elements of complex sustainable systems, in the context of the challenge of addressing societal needs within our planetary boundaries. - Evaluate processes of change towards a sustainable future at different scales, from individual users to complete societal systems. - Form arguments for your own perspective on how transitions could take place. - Work with at least three different design approaches for sustainability. - Select from a wide range of state-of-the-art design tools and techniques, introduced by experts in guest lectures and workshops. - Apply different design approaches, tools and techniques in practical projects. - Gain experience in carrying out project work in multidisciplinary teams. - Apply communication, entrepreneurship, and collaboration skills towards sustainability challenges in the real world.
Program Structure 1	Throughout the semester you will work in teams composed of students from diverse educational background. A Design Project is combined with lectures, expert talks, workshops, group discussions and field trips. There is cross fertilisation of knowledge and experiences between the courses, they cannot be taken separately. This minor consists of 5 courses: - The Design Challenge (13 ECTS, Q1&2) - Sustainability Issues and Societal Change (4ECTS, Q1) - Value Sensitive Design (4ECTS, Q1) - Masterclasses (6ECTS, Q2) - Demystify Green (3ECTS, Q2)
Administration by the Faculty of	IO
Intended for	Students from Industrial Design Engineering (IDE) and other TU Delft faculties, as well as (international) students from design and engineering backgrounds, on a level comparable to TU Delft.
Minor Content 1	What would a sustainable future look like? And how can we design for it? Developing your own answers to these two questions will be at the centre of this minor. Political leaders (agreed, not all), business leaders and members of society recognize the sustainability crisis that we are in; climate change, loss of biodiversity, the threat it poses to societies, economies and our individual well-being. This minor will help you understand the way of thinking that is needed to establish change and the part you can play in responding to the challenges in front of us and ahead of us. You will learn how to use a design approach when dealing with sustainability transitions. You will learn to use specific design approaches and methods to contribute to, facilitate or lead (?) sustainability transitions. You will be stimulated to develop your own vision on sustainability (one of the so-called wicked problems) and toolkit for sustainability that fits with your own views and values. This minor will help you improve your career opportunities: international organisations see the move towards a sustainable world as a complex, multidisciplinary challenge. The minor also collaborates on exploratory projects. You will gain unique experience working on leading topics with some frontrunner organisations and larger organisations. You will learn about different concepts that are central to designing sustainability transitions. Such as transition management, complex systems, systemic design, planetary boundaries, futuring, social design, everyday (social) practices, critical materials, worldviews, values and ethics, behaviour change, participatory design and co-creation and the circular economy.
Maximum number of participants	60
Remarks Maximum number of participants	Minors of the IDE faculty are open only for students from academic programs.

IO3834-23	Value Sensitive Design	4
Responsible Instructor	Ir. I.C. de Pauw	
Contact Hours / Week x/x/x/x		
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	When developing a new product, designers and engineers often focus on the immediate context of use: how will a product be used by the person who purchases it? They rarely have the time to consider the long-term and indirect effects of the product and their technologies. How will the use of the product affect other stakeholders? If usage of the product becomes common, how will it affect the larger community? How will people in a different culture adapt the product to their needs? What happens to the product after it is thrown away? Value sensitive design seeks to provide theory and a method to account for human values in a principled and systematic manner throughout the design process. In this course, you will learn about and practice Value Sensitive Design. The lectures address the moral, design and theoretical background of value sensitive design, and in the practicums you explore the long-term influence of new technology, using a set of Envisioning Cards developed by the University of Washington (vsdesign.org).	
Study Goals	After successfully completing the course, you are able to: - Describe the approach of Value Sensitive Design (VSD), and apply it to selected design problems. In particular, you will be able to work with the VSD envisioning cards in the design process. - Critically consider the long-term and indirect effects a product may have, from four different angles: - Stakeholders (the range of effects of a product on direct and indirect stakeholders) - Time (the long-term effects of a product after it has been appropriated in society) - Values (the impact of a product on human values) - Pervasiveness (interactions that follow from the widespread adoption of an interactive product). - Identify the (moral) values that are relevant for the design of a particular technology, and translate these values into design requirements. - Recognize value trade-offs in design, describe the various methods for dealing with value trade-offs in design and their pros and cons, and apply these methods in design. - Discuss value issues in design with others, including non-engineers.	
Education Method	The course combines theoretical work (lectures and reading assignments) with practicums. The lectures introduce you to different aspects of value sensitive design, such as stakeholder analysis, identification of values, and value conflicts. In the practicums, you will use the Envisioning Cards to explore and discuss a specific design case in groups. The first four practicums will address four dimensions of value sensitive design (stakeholders, time, values, and pervasiveness). During these four practicums, each group works on a case project, introduced in the lectures. During the fifth practicum, you will look at the integration of value sensitive design in your Minors design challenge.	
Assessment	This course has deliverables for the theoretical work and the practical work: 1) The deliverables of the theoretical work are three assignments, which consist of questions and writing tasks. These assignments will be made available through Brightspace. 2) The deliverables of the practical work are five 2-page summary reports with your main findings of each practicum, integrated in a final report. To receive your final grade, you need to submit the three individual assignments, the practicum summary reports, and the final report. Your final report and the individual assignments will be graded, resulting in a combined mark: - 3 individual assignments, together accounting to 50% of your final grade - the final practicum report accounting to 50% of your final grade. You need to complete the individual and the groups assignment with a partial grade of at least 5.	
Enrolment / Application	This course is part of the Minor Designing Sustainability Transitions (IO-MI-222), and can only be followed as part of the Minor.	

Year

Organization

Education

2024/2025

Industrial Design Engineering

Minors Industrial Design Engineering

IO-MI-223	
Minor Coordinator	K.G. Heijne
Program Title	Connected Creativity
Program Coordinator	Katrina Heijne (K.G.Heijne@tudelft.nl)
Contact for Students	connectedcreativity@tudelft.nl
Minor Remarks / Schedule	For more information about this minor see the following link: https://eduxchange.nl/for-students-tud/catalog-tud/minors/connected-creativity_54b8cb2a-777d-43d0-b178-027babb66ece

Year	2024/2025
Organization	Industrial Design Engineering
Education	Minors Industrial Design Engineering

IO-MI-240	
Minor Coordinator	E.D. van Grondelle
Contact for students	automotivedesign-io@tudelft.nl
Program Title	Future Mobility Design
EC Program	30
Introduction 1	Human mobility has evolved into a complex and multi domain challenge, which cannot be addressed in the narrow scope of a single industry. This minor is the evolution of a minor Automotive Design and anticipates the path on which the industrys scope gradually enlarges from vehicle design to a holistic mobility design approach. How to design mobility means that are meaningful within society and address both individual wants and collective needs? How to design seamless mobility, while preserving the underlying functional and emotional values of individual mobility? Fundamental questions that dictate a multidisciplinary, holistic design approach, through a context driven design process.
Introduction 2	For whom? University students from the 'designing and engineering' faculties from universities of Technology i.e., Industrial Design, Aerospace Engineering, Automotive or Mechanical Engineering and Architecture. Applications from Technology Management (TBM) are also being considered. Affinity with human mobility helps and curiosity is a must!
Introduction 3	What will you learn? The encompassing mobility domain is more complex and challenging than it ever has been, both from an internal as from an external perspective. This autonomous programme provides knowledge on automotive and mobility design and technology, and an understanding thereof in specific societal contexts. Because of the complexity of the human mobility domain and the high level of design execution in mobility-design, all knowledge, insights, and experience derived from this minor, are applicable and valuable in other complex systems and domains. You will learn how to design new meaningful products in the field of mobility. Products can be new policies and services, new vehicles in the widest possible sense, and the mobility system that incorporates all these entities. You will learn how to work with a context driven design process. Develop skills to visualize ideas appropriately throughout the design process, and to visually express all knowledge derived from that design process, through various visualization techniques. You will understand their respective roles and values in multidisciplinary design teams. Learning secondary skills like design project management and presenting your work in a professional manner, both verbally and visually. You will understand the merits and the value of design research and participate in research projects. Working in international and multidisciplinary project teams some projects, while other projects are individual. You may not always find yourself in your comfort zone.
Introduction 4	Course overview Students execute three project courses, growing in depth, width and weight, from automotive styling and technology to the design of an integrated mobility system. Those three projects together are framed by, and structured through the context driven design process Vision in Design. Together they are one full walk-through of this context driven design process. Necessary knowledge is acquired through lectures, research and skills development.
Program Structure 1	Human Mobility: The minors three project are framed by a guiding course Human Mobility, safeguarding their coherence and connection. Lectures address context driven design, design project management and program coherence. Throughout the program, students compose their own reader, their final individual deliverable. Automotive Styling: In the first project design outside-in, the emphasis is on visual qualities of a vehicle. How do people perceive and value (automotive) styling? The goal is to understand design language and the level of expertise that is required to communicate with, or fulfil a role within, the domain of vehicle design. You learn the merits of branding which you will apply on a larger scale in the next projects. Lectures include aesthetics, branding and aerodynamics. Automotive Technology: In the second design inside-out project, students design vehicles that enable people to fulfil their mobility needs in a meaningful way, in a short-term future context. Product-user interaction is investigated qualitatively as much as quantitatively. Lectures include ergonomics, safety, vehicle dynamics, technology and structures, package design and near future developments such as autonomous driving. Mobility system design: In the third Mobility Systems project, all comes together in designing a long-term future mobility system. Teams research a future context and design an integrated mobility system. After which, each team member individually designs a full vehicle within that mobility system. Ethical choices on sustainability and inclusion help the designer to take a stand. Bringing together technology, usability, and styling, elicits what the designer wants to accomplish with and for people. Lectures include the taxonomy of human goals, mobility systems (policies, services, and products), sustainability and the build environment. Automotive skills: Necessary skills, such as visualisation in the widest possible sense, are being developed in parallel in two blocks each week. One to independently (further) develop your skills, and one that addresses specifically those skills that you need in your current project.
Prerequisites Minor	Minors of the faculty of Industrial Design Engineering are open only for students from academic programmes in the third year of their Bachelor studies.
Maximum number of participants	48
Minimum number of participants	30

Ir. E.J.J. van Breemen

Unit	Industrieel Ontwerpen
Department	Emerging Materials
Telephone	+31 15 27 83437
Room	32.B-3-050

J.J.A.P.M. van Domburg

Unit	Industrieel Ontwerpen
Department	Onderwijs en Studentenzaken
Telephone	+31 15 27 83739
Room	32.C-4-110

Ir. I.C. de Pauw

Unit	Industrieel Ontwerpen
Department	Design for Sustainability
Room	32.B-3-010

ontbreekt

E.D. van Grondelle

Ir. C.P.J.M. Kroon

W.S. Elkhuisen

Ing. A.J.C. van der Helm

K.G. Heijne